SVN
ROCK ADVISORS INC., BROKERAGE

New Apartment Construction Financial Model User Manual

Evaluate and Explain the Feasibility and Profitability of New Apartment Rental Projects

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Introduction

Purpose

This user manual provides the information necessary for the user to use and modify the Excel model "New Apartment Construction" ("The Model"). The Model is intended to be used as a tool to evaluate and explain the feasibility and profitability of new apartment rental projects. The goal is to create a financial model that takes into consideration multiple parameters and assumptions relevant to apartment construction while providing consistency, transparency, flexibility and clarity.

Model Overview

There are 8 worksheets in the Apartment Rental Model, as listed below. The Model has been designed and formatted in a way that all the sheets can be directly printed or exported into PDF to letter-sized page, unless noted below. By default, all sheets are in displayed in print-area mode. User can double check that all rows and columns are included in the page prior to exporting/printing. To print/export all pages at once, please select all the sheets in Excel prior to printing/saving as PDF.

1. Cover page
2. Executive Summary
3. Schedule (formatted as 11"x17")
4. Operating Revenue and Expenses
5. Development Costs
6. Sensitivity and Assumptions Input
7. Development Cash Flow
8. 10-Yr Cash Flow IRR (formatted as 11"x17")

An individual section for each worksheet has been included below to guide the user through how to use and modify The Model in order to evaluate an apartment project.

Meaning of Different Colours

The Model has been formatted with different TEXT colours to differentiate between assumption/input cells and formula/template cells.

To maintain consistency of The Model please follow the convention below:

- BLUE – *these cells are to be changed - they are user inputs and assumptions*
- BLACK – *these cells should not be changed – they are template and formula cells.*
- GREEN - *these cells maybe changed – they are coded as formula to simplify user's work but can be changed by user to override the formula. To maintain consistency, please change the text to Blue once hard-coded.*

Similar to the Model, any inputs that need to be entered/adjusted are included as BLUE comments in this manual.

1. Cover Page

This cover page gives a quick snap shot of the project in terms of the project name, location, number of units and return summary.

Sample Apartment Development Model ← ENTER: Project Name

Toronto, Ontario ← ENTER: Project Location

245 Units

Apartment Pro-forma

Prepared by:

SVN Rock Advisors Inc. ← ENTER: Author Name

2. Executive Summary

There is no user input on this page, all information shown are pulled from all other work sheets, as discussed in subsequent sections below.

Title caries across all pages

Sensitivity Scenario on all pages

Currently running: Base Case

Sample Apartment Development Model
Toronto, Ontario
Executive Summary

)perating Income @ Stabilization				4,067,876	4,067,876	4,067,876
alization Rate				5.50%	5.00%	4.50%
ation @	$	1.90	per sqft average rent	73,961,374	81,357,511	90,397,235

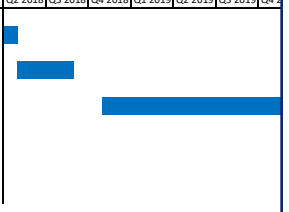
3. Schedule

This page is used to set the schedule of the project from land purchase to stabilization. A visual timeline based on the schedule is updated automatically based on the dates. Construction and lease-up durations are drawn from the sensitivity page below (Section 6.0).

Construction time, lease-up, etc. vary tremendously and will be dependent on the type of product being built, the location of the property, and the demographics of the tenants. Pre-leasing is possible three to six months prior to completion of construction.

Project Start Date **Apr-18** ← ENTER: Project Start Date

	Duration	Start Date	End Date	Anchored to	Offset
Land Purchase	0	Apr-18	Apr-18	Project Start Date	0
Pre-Development	3	May-18	Aug-18	Land Purchase	0
Construction	18	Nov-18	May-20	Pre-Development	2
Lease-up	16	Dec-19	Apr-21	Construction	-6
Stabilized	0	May-21	May-21	Lease-up	0



ENTER: Duration in months

ENTER: Gaps in months between each phase

4. Operating Revenue and Expenses

ENTER: Parking stall details

Parking		
Residential Stalls	190	
Visitor Stalls	60	
Extra Stalls	30	
Total	280	Ratio 1.14 stall per unit

ENTER: Unit and Rent details

Revenue							
Unit Type	Plan	Sqft	No. Units	% Total	Monthly $	Annual $	$/Sqft
1 Bedroom	A	600	60	24%	1,200	14,400	2.00
1 Bedroom + Den	B	700	30	12%	1,300	15,600	1.86
2 Bedroom	C	800	80	33%	1,500	18,000	1.88
2 Bedroom + Den	D	900	35	14%	1,600	19,200	1.78
3 Bedroom	E	1,150	30	12%	1,700	20,400	1.48
3 Bedroom + Den	F	1,250	10	4%	2,000	24,000	1.60
Total		199,500	245	100%	358,000	4,296,000	
Average		814			1,461	17,535	1.79

ENTER: Retail rent and areas

ENTER: $/stall rent

Other Income					
Retail Revenue	$20 /sqft	x	3,000 sqft	60,000	720,000
Other Revenue				-	-
Parking Revenue	190 stalls	x	$100	19,000	228,000
Total Other Income				79,000	948,000
Total Potential Gross Revenue				437,000	5,244,000
Less: Vacancy and collection loss		@	3.0%	13,110	157,320
Effective Gross Income				423,890	5,086,680

ENTER: % vacancy

ENTER: Expenses per unit/ year

Operating Expenses		Per Unit	%	Per Sqft $	Monthly $	Annual $
Property Taxes		1,500	29%	1.84	30,625	367,500
Insurance		150	3%	0.18	3,063	36,750
Repair & Maintenance		850	17%	1.04	17,354	208,250
Heat		240	5%	0.29	4,900	58,800
Hydro		500	10%	0.61	10,208	122,500
Water		400	8%	0.49	8,167	98,000
Staffing		750	15%	0.92	15,313	183,750
Marketing		100	2%	0.12	2,042	24,500
Miscellaneous/ Office		100	2%	0.12	2,042	24,500
Management Fee	3%	526	10%	0.65	10,740	128,880
Total Expenses		5,116	100%	6.3	104,453	1,253,430
Operating Expenses Varience						
Operating Expenses						1,253,430
Operating Expenses/Revenue						25%

Net Operating Income and Valuation						
NOI Today	@	$ 1.79	*per sqft average rent*		$	3,833,250
Valuation today	@	5.00%	Cap rate	312,918 per unit	**Total $**	**76,665,000**
NOI at Stabilization	@	$ 1.90	*per sqft average rent*		$	4,067,876
Valuation @ Stabilization	@	5.00%	Cap rate		**Total $**	**81,357,511**

ENTER: Management fee %

Op. Exp. Variance - from sensitivity page

Revenue

In order to determine **potential rental revenue**, rents to be charged can be projected by considering the following data in this order:

1. Any newly constructed buildings in the property's catchment area or a similar catchment area.
2. Condo rentals in the area. While these are new construction in terms of quality, the marking positioning in the market, and tenure are not equivalent to a new purpose-built rental building.
3. Rents being charged in older apartment buildings in the area that have had a full renovation (kitchens, bathrooms, etc.).
4. Rents being charged in conventional apartments close to the property (without renovations).
5. CMHC average rents as a broad indicator of the market.

Parking income is dependent on the location, and will be determined based on a market survey.

Vacancy rates vary depending on the municipality and rent level.

Operating Costs

Realty Tax: realty tax can vary tremendously depending on the municipality. Determining municipal taxes is complex. At a high level, realty taxes can be estimated by creating a blanket per unit valuation based on the location, scale, and quality of the proposed development, and applying the municipal mill rate. We can provide assistance with determining realty taxes and appealing for lower assessment values.

Repairs & Maintenance: an industry-accepted number used for newly constructed rentals is $850 per unit annually. Actual costs vary depending on the level of maintenance required given building features and amenities.

Utilities: utility costs will vary depending on the HVAC system in the building and the municipality.
Assuming central heat/chilling, water, and common area electricity, and energy efficient building features, costs for utilities can be expected to be $1,400 per unit annually. This amount covers heat, electricity and water, each accounting for approximately 17%, 43%, and 40%, respectively. Electricity rates may fluctuate depending on current Provincial policies and rebates offered. You will require professional advice on utility costs as you develop your proforma and your building HVAC system.

Staffing/Security: an industry standard for high quality rental apartment staffing is $750-$1000 per unit annually. This amount assumes there is no 24-hour concierge provided. If a concierge is provided at the property, staffing expenses would rise significantly.

Marketing: marketing can vary tremendously, we have used a cost of $100/unit annually, which is reasonable upon stabilization, but not during initial lease-up.

Insurance: insurance rates will vary widely, we have used $150 per unit annually.

Property Management: fees for property management generally range from 2.5%-5.0% of gross revenues for the property, we have used a cost of 3.0% of gross revenues for the property.

5. Development Costs

Input the costs for the development on this sheet. Although all the information is contained on one work sheet, this sheet will be exported and printed on 2 separate pages, given the space. Development costs are typically provided by the developer or a quantity cost surveyor. The user can change headings and leave costs blank as required.

First page contains: land, construction, professional fees and development management fees.

Sample Apartment Development Model
Toronto, Ontario
Development Costs

ENTER: Total costs for each line item

ENTER: % contingency for each cost category

ENTER: Development management fee

			Total	Per unit	Per Sqft
Number of Units:				245	
Net Rentable Area:				199,500	
LAND					
Land			20,000,000	81,633	100.3
Closing Cost		1%	200,000	816	1.0
Total Land Costs			20,200,000	82,449	101.3
CONSTRUCTION					
Construction hard cost			35,000,000	142,857	175.4
Offsite/Sitework Costs			-	-	-
Furniture Fixtures & Equipment			-	-	-
Construction contingency		3%	1,050,000	4,286	5.3
Total Construction Costs			36,050,000	312,041	383.2
PROFESSIONAL FEES					
Architect			100,000	408	0.5
Landscape architect			20,000	82	0.1
Structural engineer			50,000	204	0.3
Civil engineer			50,000	204	0.3
Electrical engineer			50,000	204	0.3
Geotechnical engineer			40,000	163	0.2
Mechanical engineer			100,000	408	0.5
Survey			50,000	204	0.3
Building envelope consultant			50,000	204	0.3
Environmental consultant			5,000	20	0.0
Interior design			40,000	163	0.2
Traffic consultant			5,000	20	0.0
Other consultant			50,000	204	0.3
Disbursements			50,000	204	0.3
Professional fee contingency		5%	33,000	135	0.2
Total Professional Fees			693,000	2,829	3.5
DEVELOPMENT MANAGEMENT					
Construction			36,050,000		
Professional fees			693,000		
Permit and approvals			1,323,000		
Marketing			630,000		
			38,696,000		
Development management fees	@	2%	773,920	3,159	3.9

Second page contains: permits and approvals, marketing and leasing, financing and Interest, and operating income during lease-up.

ENTER: Total costs for each line item

ENTER: % contingency for each cost category

ENTER: Property tax rate (for reference only), total hard coded

			Total	Per unit	Per Sqft
PERMITS AND APPROVALS					
Development permit			10,000	41	0.1
Rezoning			-	-	-
Building permit			200,000	816	1.0
Other development charges			300,000	1,224	1.5
Park Dedication Fee			50,000	204	0.3
City Development Charges			500,000	2,041	2.5
Utilities Connection			100,000	408	0.5
Land titles			100,000	408	0.5
Permit and approval contingency		5%	63,000	257	0.3
Total Permit and Approvals			1,323,000	5,400	6.6
MARKETINGAND LEASING					
Marketing			100,000	408	0.5
Leasing commissions			300,000	1,224	1.5
Leasing concessions			200,000	816	1.0
Miscellaneous			-	-	-
Marketing and leasing contingency		5%	30,000	122	0.2
Total Marketing and Leasing			630,000	2,571	3.2
FINANCING AND INTEREST					
Appraisal			20,000	82	0.1
Legal fees			80,000	327	0.4
Property Taxes During Construction	Tax rate	1.73%	200,000	816	1.0
Quantity Surveyor			300,000	1,224	1.5
Insurance			50,000	204	0.3
LC and Bonding			500,000	2,041	2.5
Loan fees			100,000	408	0.5
Bank charges			100,000	408	0.5
HST			1,500,000	6,122	7.5
Construction Loan Interest	@	5%	4,083,682	16,668	20.5
Financing contingency		5%	142,500	582	0.7
Total Financing and Interest Costs			7,076,182	28,882	35.5
OPERATING INCOME DURING LEASE-UP					
Revenue			(4,187,538)	(17,092)	(21)
Vacancy Provision			125,629	513	1
Operating Expenses			1,000,937	4,085	5
Total Income During Lease-up			(3,061,072)	(12,494)	(15)
TOTAL DEVELOPMENT COSTS			**63,685,029**	**259,939**	**319**

Construction loan interest rate - from sensitivity page

6. Sensitivity and Assumptions Input

Generally standard parameters are included in the Model and screenshot below, but these can be revised to fit the expected timelines and characteristics of the project. Capitalization rates will be very dependent on the market where the property is located.

Select Scenario: Base Case

Select scenario from the drop-down menu to analyze sensitivity. This is set to Base Case by default.

Capitalization Rate	**5.00%**
Base Case	5.00%
Best Case	4.50%
Worst Case	5.50%
Construction Loan Interest Rate	**4.5%**
Base Case	4.5%
Best Case	4.0%
Worst Case	5.0%
Construction Period (Months)	**18**
Base Case	18
Best Case	16
Worst Case	20
Lease-up Period (Months)	**16**
Base Case	16
Best Case	12
Worst Case	20
Operating Expense Varience	**0.0%**
Base Case	0.0%
Best Case	-5.0%
Worst Case	5.0%

ENTER: Parameters for each sensitivity scenario. If no variation is expected for a parameter, then input the same value for all three scenarios.

7. Development Cash Flow

This sheet includes a month-by-month development cash flow of the project from start to rent stabilization. There are many components to this page as described in each subsection below.

7.1 Lease-Up Schedule

Dates tabulate automatically based on Schedule page

Items	Budget	Variance	Total Cashflow	Start	End	Dec 19	Jan 20	Feb 20
LEASE-UP SCHEDULE								
Number of Units Completed	245	-	245			-	-	245
Cumulative Units Complete	245	-	245			-	-	245
Percentage Completed			100%			0%	0%	100%
New Units Leased	245	-	245	Sep-19	Jan-21	14	14	14
Cumulative Units Leased	245	-	245			58	72	86
Percentage Leased			100%			24%	29%	35%
Units Occupied	245	-	245			-	-	86
Cumulative Units Occupied	245	-	245			-	-	86
Percentage Occupied			100%			0%	0%	35%

From Schedule page, by default model assumes same number of units leased per month, can be updated manually.

Units completed based on construction from Schedule page. Can be input manually.

7.2 Development Cost Cash Flow

Use drop-down menu to select construction cost distribution over construction period. This is set to S Curve by default.
Note: unhide rows 31-37 to adjust steepness of S Curve. (This step is optional).

This column is used as a check to verify that total cash flow adds up to the same amount as the budget (should be 0).

LAND							
Land	(20,000,000)	-	(20,000,000)	Jan-18	Jan-18	-	-
Closing Cost	(200,000)	-	(200,000)	Jan-18	Jan-18	-	-
Total Land Costs	(20,200,000)	-	(20,200,000)			-	-
S Curve Distribution						4%	8%
S Curve Distribution	(40,000,000)	-	(40,000,000)	Aug-18	Feb-20	(1,570,766)	(1,725,746)
Even Distribution	-	-	-	Aug-18	Feb-20	-	-
Furniture Fixtures & Equipment	-	-	-	Aug-18	Feb-20	-	-
Construction contingency	(1,200,000)	-	(1,200,000)	Aug-18	Feb-20	(63,158)	(63,158)
Total Construction Costs	(41,200,000)	-	(41,200,000)			(1,633,924)	(1,788,904)
PROFESSIONAL FEES							
Architect	(100,000)	-	(100,000)	Feb-18	Feb-20	(4,000)	(4,000)
Landscape architect	(20,000)	-	(20,000)	Feb-18	Feb-20	(800)	(800)
Structural engineer	(50,000)	-	(50,000)	Feb-18	Feb-20	(2,000)	(2,000)
Civil engineer	(50,000)	-	(50,000)	Feb-18	Feb-20	(2,000)	(2,000)
Electrical engineer	(50,000)	-	(50,000)	Feb-18	Feb-20	(2,000)	(2,000)
Geotechnical engineer	(40,000)	-	(40,000)	Feb-18	Feb-20	(1,600)	(1,600)
Mechanical engineer	(100,000)	-	(100,000)	Feb-18	Feb-20	(4,000)	(4,000)
Survey	(50,000)	-	(50,000)	Feb-18	Feb-20	(2,000)	(2,000)

These line items are imported from Development Cost sheet.
If more items need to be added, please adjust both Cash Flow and Development Cost sheets.

From Schedule page, by default. Enter the start and end date of each activity and cash flow will update accordingly.

7.2 Development Cost Cash Flow

HST $ distributed according to unit completion from lease-up schedule

Interest calculated based on monthly construction loan balance.

Maximum construction loan balance.

HST	(1,500,000)	-	(1,500,000)				-	-
Construction Loan Interest	(3,294,320)	-	(3,294,320)				-	-
Financing contingency	(142,500)	-	(142,500)	Feb-18	Feb-20	25	-	(5,700)
Total Financing and Interest Costs	(6,286,820)	(533,333)	(5,753,486)				(33,333)	(69,033)
CONSTRUCTION LOAN								
Cash Flow before loan draws	(68,393,114)	(976,667)	(67,416,448)				(20,303,908)	(195,848)
Construction Loan Draws			43,478,858				-	-
Repayment of Construction Loan			(43,478,858)				-	-
Valuation at stabilization								-
Selling Costs							-	-
Net Cash Flow - Equity	(68,393,114)	(976,667)	(67,416,448)			-	(20,303,908)	(195,848)
				IRR	8.8%			
Cumulative		Loan to Cost:						
Construction Loan Balance	(44,024,520)	65%						-
Equity	(23,937,590)	35%						20,499,756
Development Status							PUD	PUD
Cost to Date							(20,303,908)	(20,499,756)
Cost to Complete Financing		-	(23,937,590)				-	-

IRR upon stabilization, if sold

ENTER: Loan to Cost % for construction loan.

Development Status: “PUD” from start to stabilization, “SLD” after that.

Goal seek to 0 by changing

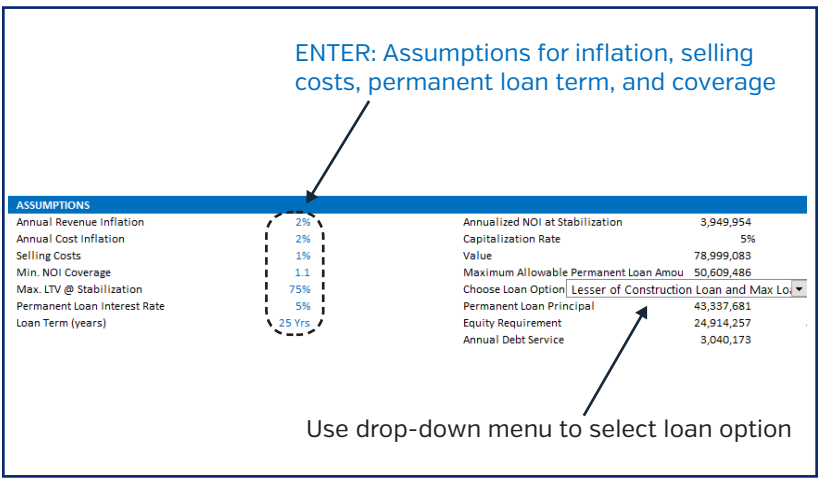
Go to Tab “Data” > What-If Analysis > Goal seek to 0 by changing cell indicated. *This method is used to calculate equity requirement while avoiding circular references from interest calculation*

8. 10-Year Cash Flow IRR

Analysis of project permanent financing, cash flow, and IRR from year 1-10 are included in this sheet, and the timeline for IRR analysis can be revised as needed.

8.1 Assumptions and Permanent Financing Option

Financing varies depending on the project and the developer's financial position, please contact us to discuss these input values further.



8.2 Cash Flow and IRR

Outstanding loan balance in each year

YEAR	1	2	3	4	5
	2018	2019	2020	2021	2022
Construction Loan Balan	7,554,210	38,410,392	43,745,722	(976,667)	(976,667)
Permanent Loan Balance	-	-	-	43,478,858	42,734,903
Stabilized NOI	-	-	-	3,137,223	3,839,961
Interest Payment	-	-	-	(1,797,776)	(2,115,521)
Prinpical Payment	-	-	-	(743,954)	(934,556)
Income before Tax	-	-	-	595,492	789,884
Valuation @ 5.50% Cap r	-	-	-	68,448,495	69,817,465
Less: Selling Cost @ 1.00%	-	-	-	(684,485)	(698,175)
Less: Loan Balance	-	-	-	(43,478,858)	(42,734,903)
Net Proceeds from Sale	-	-	-	24,285,152	26,384,387
Valuation @ 5.00% Cap r	-	-	-	75,293,344	76,799,211
Less: Selling Cost @ 1.00%	-	-	-	(752,933)	(767,992)
Less: Loan Balance	-	-	-	(43,478,858)	(42,734,903)
Net Proceeds from Sale	-	-	-	31,061,553	33,296,316

Forecasted net proceeds for sale in each year, based on different cap rate scenarios.

YEAR	1	2	3	4	5	6
	2018	2019	2020	2021	2022	2023
IRR CALCULATION	*Based on valuation @ 5.00% Cap rate*					
Equity Investment	(23,937,590)	-	-	(976,667)	-	-
Income before Tax	-	-	-	595,492	789,884	866,683
5 -year cash flo	(23,937,590)	-	-	(381,175)	34,086,199	-
IRR 8.9%						
7 -year cash flo	(23,937,590)	-	-	(381,175)	789,884	866,683
IRR 9.2%						
10 -year cash flo	(23,937,590)	-	-	(381,175)	789,884	866,683
IRR 9.2%						

ENTER: year #'s for IRR analysis based on selling building in those years.

The third option is included on the executive summary and cover pages.

USB Contents

The USB contains:

1. The complete, functional *New Apartment Construction Financial Model* workbook.
2. The user manual.
3. Samples of PDF version of compete *New Apartment Construction Financial Model*.

Disclaimer

By purchasing and making use of the product, the user agrees and acknowledges that use of the unlocked financial Model is for the user's internal financial analysis only.

For greater certainty, the user is in no circumstances permitted to share the unlocked excel file with any other parties, including its representatives, lenders, advisors, etc.

Only locked PDF copies of the Model may be shared with parties other than the initial purchaser and its immediate employees.

Intellectual rights to the Model remain with SVN Rock Advisors Inc., Brokerage unless agreed otherwise in advance.

Please contact SVN Rock Advisors Research and Consultancy Department for additional services for your new apartment construction development project.

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